

Extending RF Fingerprinting-Based Physical Layer Authentication Through Additional Machine-Learning Models for Unauthorized Device Detection

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Abstract

This project evaluates Physical Layer Authentication (PLA) for detecting unauthorized wireless transmitters using radio-frequency fingerprints. It follows the RF Fingerprinting project option proposed in the course and uses the public PLA implementation referenced in the lecture slides. The supplied framework derives fingerprints from I/Q signals affected by device-dependent hardware imperfections and classifies legitimate and rogue devices using machine learning. The implemented contribution is to evaluate two additional classifiers, Gaussian Naive Bayes and Extra Trees, while retaining the available Discrete Gabor Transform (DGT)-derived features and the three official authorized-versus-roguer scenarios. Under a repository-compatible ANOVA comparison, Gaussian Naive Bayes obtains a mean Average Detection Rate (ADR) of 89.83% and Extra Trees obtains 83.56%; neither outperforms the supplied Random Forest baseline of 98.38%. A secondary leakage-safe stress test, where rogue signatures are excluded from training and candidate selection, reveals substantially lower legitimate-device authentication performance. The results show both the effectiveness of the supplied baseline and the need for strict evaluation against genuinely unseen impersonation attempts.

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1 Introduction

Wireless-device authentication is a key security requirement in Internet of Things (IoT), edge and next-generation wireless networks. In these systems, a malicious transmitter may attempt to impersonate a trusted node or inject traffic while appearing to originate from an authorized device. Cryptographic authentication remains essential, but physical-layer techniques can provide an additional security signal based on device-specific transmission characteristics.

The selected course project is explicitly based on the professor’s RF Fingerprinting proposal: the Physical Layer Authentication (PLA) repository is provided, the existing dataset is based on I/Q signals collected from BladeRF with injected hardware imperfections, the signal-processing pipeline uses the Discrete Gabor Transform (DGT), and students may compare alternative machine-learning models with those proposed in the paper (Loscri, 2026a). Accordingly, this project keeps the provided DGT-derived fingerprint features and concentrates on a reproducible ML-model extension.

2 Background and Security Context

2.1 Physical Layer Authentication

PLA authenticates a radio transmitter using characteristics observed in its signal rather than relying only on identifiers or application-layer credentials. The referenced PLA approach uses inherent radio hardware impairments as discriminative fingerprints. The official repository and lecture presentation identify three relevant impairments:

- **Carrier Frequency Offset (CFO):** mismatch between the transmitter carrier and the receiver’s expected carrier frequency;
- **Direct Current Offset (DCO):** unwanted DC components affecting the in-phase and quadrature components;
- **Phase Offset (PO):** phase deviation arising from hardware sources such as oscillator and phase-locked-loop behaviour.

The repository describes the method as an ML-assisted PLA approach for authenticating devices and detecting unauthorized access through CFO, DCO and PO fingerprints (PLA-AP, 2026; Alla et al., 2024).

2.2 DGT-based feature generation

The course material shows a DGT-based processing chain: the input signal is transformed into Gabor coefficients, RF signature regions are segmented, and statistical descriptors such as standard deviation, variance, skewness, kurtosis and entropy are extracted. These derived features form the input to machine-learning authentication models (Loscri, 2026b).

2.3 Threat model

The attack studied in this project is **device impersonation**. An unauthorized transmitter presents radio emissions to the authentication mechanism and attempts to be accepted as the legitimate target device. For a target device under verification, the binary labels are:

$$y = \begin{cases} 1, & \text{target authorized device,} \\ 0, & \text{other authorized or rogue device.} \end{cases} \quad (1)$$

The defender must accept genuine target-device transmissions while rejecting rogue signatures. The lecture presentation specifies that malicious RF signatures are to be used only for testing rather than for training or model selection, to evaluate unseen unauthorized-device detection (Loscri, 2026b).

3 Research Question and Project Contribution

3.1 Research question

Can additional machine-learning classifiers improve unauthorized-device detection in the DGT-based PLA RF-fingerprinting framework across the supplied authorized-versus-rogue scenarios?

3.2 Chosen contribution

The official repository supports Random Forest, SVC, KNN, XGBoost and Logistic Regression as executable classifier options, while the supplied result files additionally include Gradient Boosting outputs. The project extends the comparison with:

- **Gaussian Naive Bayes (GaussianNB):** a lightweight probabilistic classifier that tests whether individual selected features provide adequate separability;
- **Extra Trees Classifier:** a randomized tree ensemble that tests whether stronger tree randomization improves robustness over the existing Random Forest results.

A second implementation also evaluates Linear Discriminant Analysis (LDA) under a stricter rogue-held-out protocol as an additional sensitivity analysis.

4 Dataset and Experimental Setup

4.1 Available processed dataset

The official uploaded repository contains `dataset/processed_fingerprints_data.h5`, which was inspected and used for the completed experiments. Table 1 summarises the usable processed dataset.

Table 1: Inspected processed RF fingerprint dataset.

Property	Value
Total processed samples	146
DGT-derived features per sample	505
Transmitting device labels	12
Samples per device	6–17

The larger raw and AWGN-noise datasets were not available in the current environment due to their size. This does not prevent the primary classifier-comparison experiment, as the official repository states that the processed HDF5 fingerprints are used for training and validation (PLA-AP, 2026).

4.2 Official trial scenarios

The same three scenario definitions distributed with the official PLA implementation were retained. They progressively change the ratio of authorized to rogue devices.

Table 2: Official authorized-versus-rogue device scenarios.

Scenario	Authorized devices	Rogue devices
Trial 1	4	8
Trial 2	6	6
Trial 3	8	4

4.3 Metrics

The course slides specify target criteria of authorized True Detection Rate (TDR) of at least 95% and authorized False Detection Rate (FDR) below 5%. The official PLA evaluation also uses a malicious-device detection rate. In this project the principal aggregate value is Average Detection Rate:

$$ADR = \frac{TDR_{authorized} + TDR_{rogue}}{2}. \quad (2)$$

Here, $TDR_{authorized}$ represents correct acceptance of the target legitimate device and TDR_{rogue} represents correct rejection of rogue transmitters.

5 Implementation and Evaluation Protocols

5.1 Repository-compatible comparison

The first experiment compares the newly added classifiers against the official ANOVA result files. ANOVA was selected because it is explicitly included in the supplied framework and is considerably faster to execute than RFE and several alternative combinations reported in the repository documentation (PLA-AP, 2026).

The added-model script uses candidate feature counts of 10, 50, 100, 200 and 505, then follows the distributed model-selection style closely enough to compare its output to the existing saved results. The principal command is:

```
python RF_Fingerprint_Repository-Compatible_Extension.py
```

5.2 Security caveat and leakage-safe stress test

Inspection of the distributed evaluation logic showed that rogue-device outcomes are involved when selecting a best candidate configuration. That method is suitable for reproducing or extending the saved-result style, but it provides an optimistic security estimate when the deployment goal is detection of previously unseen attackers. This is particularly important because the lecture material expressly states that malicious signatures should be reserved for testing.

A secondary script therefore applies a stricter protocol:

- ANOVA feature selection is fitted only on authorized training folds;
- 50 selected features are fixed without observing rogue outcomes;
- rogue signatures are used only as an external impersonation test set;
- results are reported separately from the repository-compatible comparison.

The command is:

```
python RF_Fingerprint_Extended.py --data dataset/processed_fingerprints_data.h5 \
--results-dir results/strict_rogue_held_out \
--figures-dir figures/strict_rogue_held_out
```

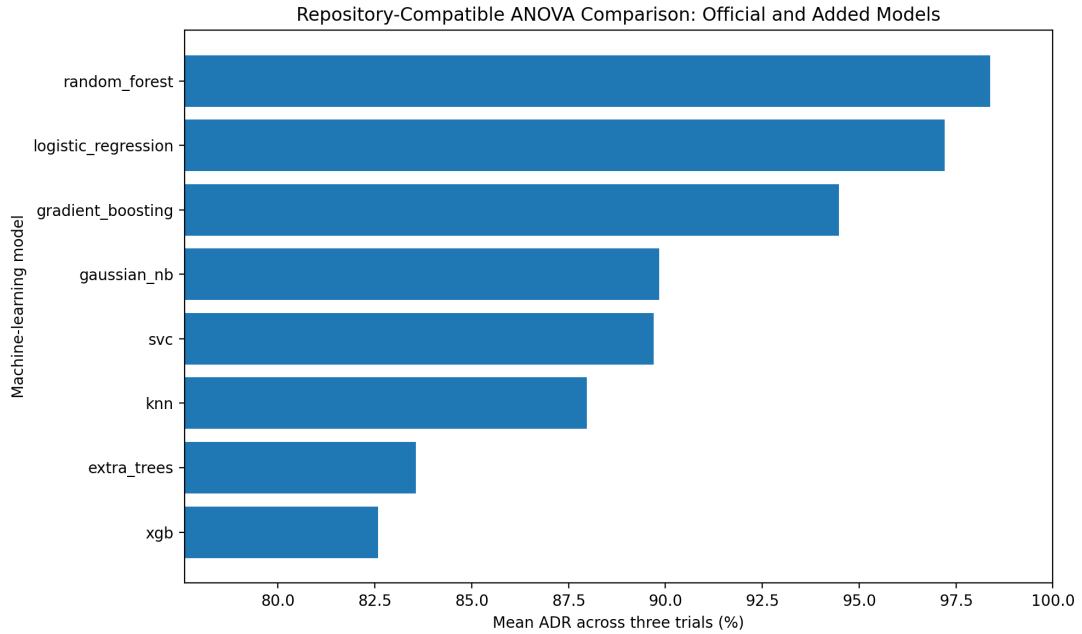
6 Results

6.1 Repository-compatible ANOVA model comparison

Table 3 reports the principal result: the supplied ANOVA baselines compared with the two newly executed classifiers over all three scenarios.

Table 3: Repository-compatible ANOVA comparison across the three official trials.

Model	Source	Mean ADR
Random Forest	Supplied baseline	98.38%
Logistic Regression	Supplied baseline	97.21%
Gradient Boosting	Supplied baseline	94.48%
Gaussian Naive Bayes	Added and executed	89.83%
SVC	Supplied baseline	89.70%
KNN	Supplied baseline	87.96%
Extra Trees	Added and executed	83.56%
XGBoost	Supplied baseline	82.58%

**Figure 1:** Mean ADR under the repository-compatible ANOVA evaluation. The strongest supplied baseline remains Random Forest.

The central finding is that neither added classifier improves on the strongest supplied model. GaussianNB slightly exceeds the supplied SVC result but remains substantially below Random Forest and Logistic Regression. Extra Trees does not outperform Random Forest, suggesting that additional tree randomization is not beneficial for this available processed dataset and selection procedure.

6.2 Stricter rogue-held-out security stress test

Table 4 shows the second experiment. These values must not be interpreted as replacements for the supplied baseline because the protocol is deliberately stricter: rogue outcomes cannot influence feature selection or model configuration.

Table 4: Leakage-safe rogue-held-out stress test using fixed ANOVA feature selection.

Model	Category	Authorized TDR	Rogue rejection	Mean ADR
LDA	Added	23.10%	85.68%	54.39%
Logistic Regression	Existing	31.74%	74.74%	53.24%
Extra Trees	Added	4.39%	96.91%	50.65%
Random Forest	Existing	3.06%	97.42%	50.24%

The strict test exposes a practical security trade-off. Random Forest and Extra Trees reject almost all rogue transmissions, but in this setting they also reject nearly all genuine target transmissions. LDA gives the highest ADR in the strict experiment, yet its legitimate-device TDR remains too low for deployment. The test therefore indicates that the small processed dataset is insufficient for claiming robust generalisation against unseen impersonation attempts under this conservative protocol.

7 Noise Robustness Evidence and Limitation

The official repository includes saved results for AWGN/SNR experiments at 1, 3, 5, 7, 10 and 15 dB, consistent with the SNR range presented in the lecture material. Figure 2 visualises those supplied summary results.

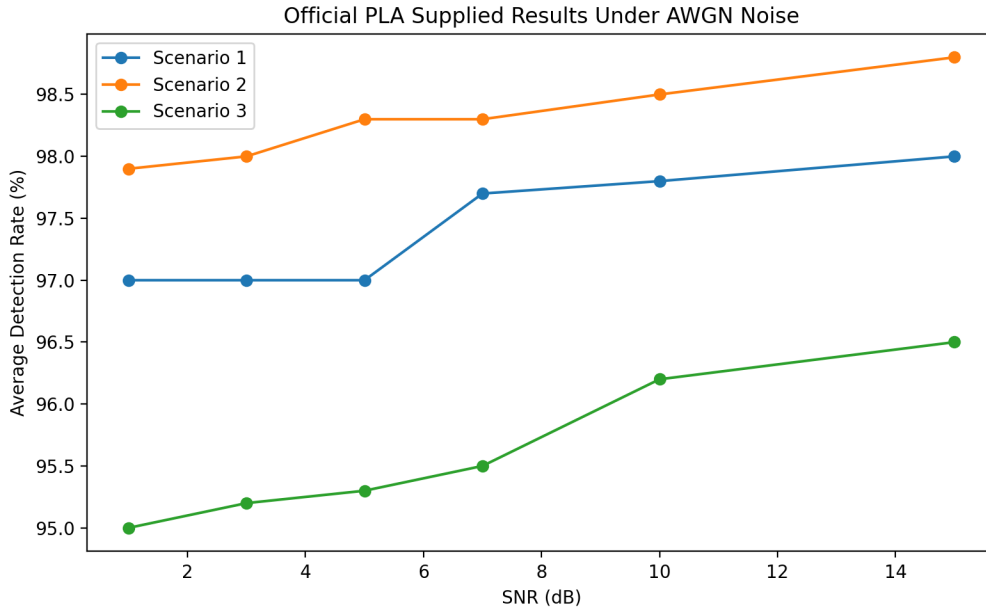


Figure 2: Official supplied noise-result summaries over available SNR values. These are not newly executed added-model results.

The full noise dataset is approximately 8 GB and was not available for execution in the current working environment. Consequently, this report does not claim that GaussianNB, Extra Trees or LDA were tested under official AWGN conditions. A future local experiment can execute the added models after downloading the full noise dataset onto a machine with sufficient storage.

8 Security Analysis and Discussion

8.1 Interpretation of the primary result

The professor-approved extension has been completed: additional ML methods were integrated and compared against supplied original baselines using the official processed fingerprints. The outcome is academically meaningful even though the new methods do not win. Random Forest remains the strongest supplied ANOVA baseline, which suggests that the data representation and sample regime favour its nonlinear ensemble structure.

8.2 Evaluation realism

A central security lesson is that attacker samples must not be allowed to influence model selection when claiming detection of unseen attackers. A method that selects features or configurations based on observed rogue outcomes is measuring performance on known attack signatures rather than generalisation to new impersonators. The separated stress test avoids this issue and demonstrates that performance is much less optimistic in the available small-sample setting.

8.3 Limitations

- The available processed file has only 146 observations across 12 devices; some devices contain as few as six samples.
- The raw I/Q acquisition process and DGT feature-generation stage were not regenerated, because the current experiment focuses on the permitted ML-comparison extension.
- The full AWGN noise dataset was not executed for new classifiers due to its size.
- The repository-compatible figures are useful for comparison with supplied outputs, but should not alone be treated as a leakage-safe deployment estimate.

8.4 Recommended countermeasures and future work

A robust deployment-oriented PLA evaluation should isolate attacker devices from all training and selection decisions; expand the number of transmitters and samples per transmitter; evaluate changing SNR, frequency channel and receiver conditions; and report both unauthorized rejection and legitimate acceptance so that a model is not rewarded merely for rejecting all signals.

9 Conclusion

This project implements the RF Fingerprinting / PLA option proposed in the course and directly uses the official repository and processed fingerprint dataset. The selected contribution – comparing additional machine-learning classifiers – was completed with real executions of Gaussian Naive Bayes and Extra Trees over the three supplied trial scenarios. In the repository-compatible ANOVA comparison, GaussianNB achieved 89.83% mean ADR and Extra Trees achieved 83.56%, while the supplied Random Forest baseline remained best at 98.38%.

The supplementary rogue-held-out stress test contributes a more critical security perspective: high rogue rejection does not imply deployable authentication if legitimate transmitters are rejected at high rates. The project therefore provides both a completed ML extension and a grounded discussion of how evaluation methodology affects security claims in RF fingerprinting systems.

A Reproduction Commands

The implemented project package contains the processed dataset, scripts, generated figures and CSV outputs. From the project directory, the principal commands are:

```
# Repository-compatible extension experiment
python RF_Fingerprint_Repository-Compatible_Extension.py

# Strict rogue-held-out stress test
python RF_Fingerprint_Extended.py \
  --data dataset/processed_fingerprints_data.h5 \
  --results-dir results/strict_rogue_held_out \
  --figures-dir figures/strict_rogue_held_out
```

B Submitted Files

File	Purpose
RF_Fingerprint_Repository-Compatible_Extension.py	Adds GaussianNB and Extra Trees for the main comparison.
RF_Fingerprint_Extended.py	Implements the stricter rogue-held-out stress test.
results/repository_compatible_added_model.csv	Principal CSV and JSON results.
results/strict_rogue_held_out/	Security stress-test result files.
figures/	Generated result figures.

References

- I. Alla, S. Yahia, V. Loscri, and H. Eldeeb. Robust Device Authentication in Multi-Node Networks: ML-Assisted Hybrid PLA Exploiting Hardware Impairments. In *Annual Computer Security Applications Conference (ACSAC)*, pp. 1172–1185, 2024.
- PLA-AP. Physical Layer Authentication (PLA) Approach Repository. <https://github.com/PLA-AP/PLA>, accessed 26 May 2026.
- V. Loscri. Module 1 – Introduction to Advanced Network Security: Examples of Projects – RF Fingerprinting. Course slides, *Security of Advanced Networking and Services*, p. 89, 2026.
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